Probabilistic Reasoning

A brief review of Probability

From Bernoulli to Multinomial Distribution

Bayes Classification – Bayes Theorem

Naïve Bayes Classification – Attributes independence

Joint Distributions, independence, conditional independence

Markov Modal

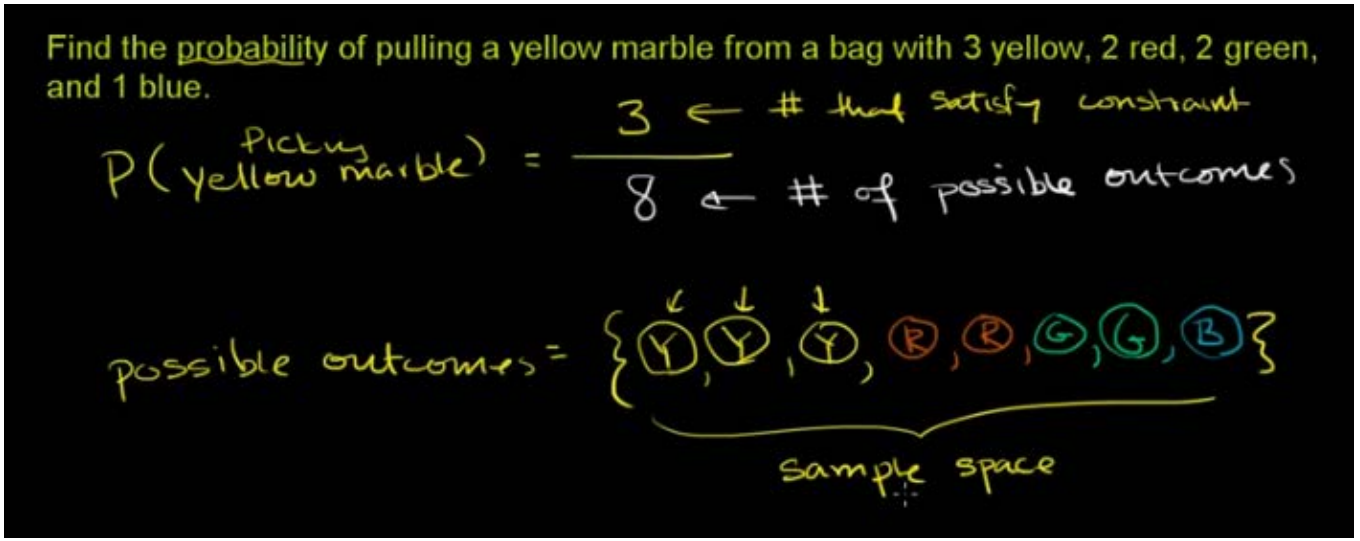
Baysian Network

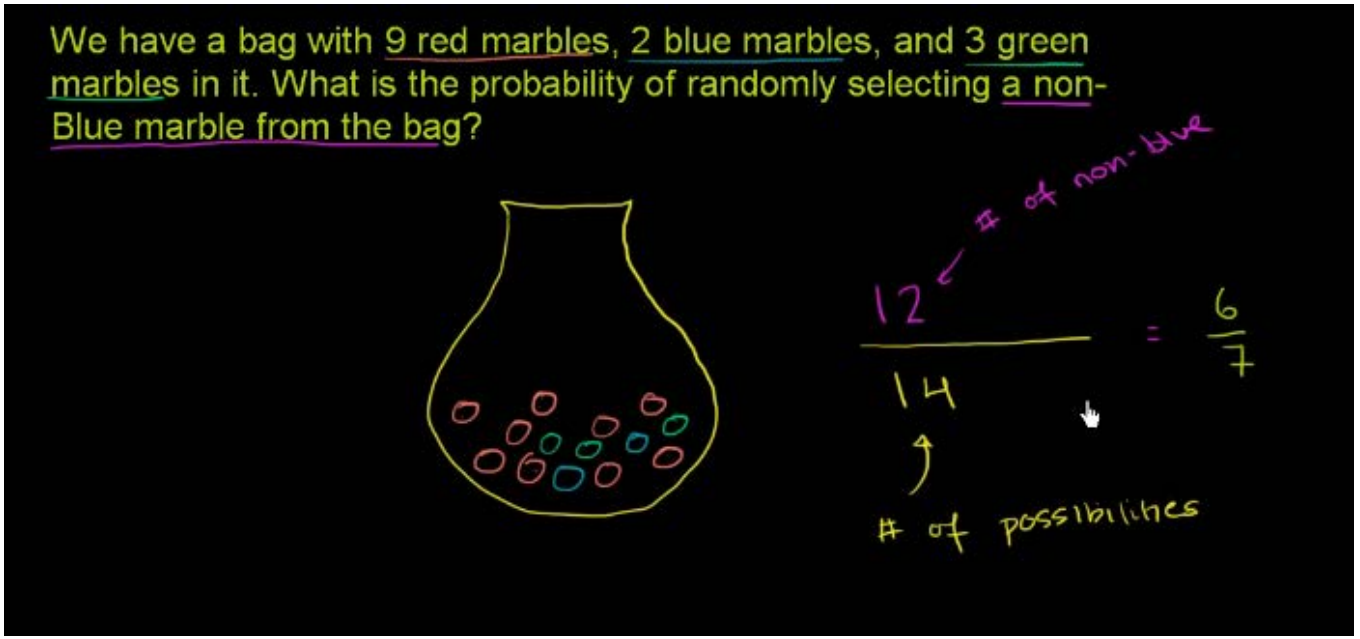
Dr. Chris Ding

From Bernoulli to Multinomial Distribution

- A brief review of probability, Bernoulli distribution, binomial distribution and multinomial distribution.
- Multinomial distribution plays vital important role in data mining/machine learning:
  - The basic model of English text, documents, fundamental theory for information retrieval, search engine, etc.
  - The basis for logistic regression and neural networks.
- An alternative (better) model of Naïve Bayes Classification

Probability = count possible outcomes satisfying requirements/constraints





How many times people eat food every day ?

Many people will say 3 times, breakfast, lunch, dinner

A coin has 2 sides, head and tail side.

If you throw 10 coins up and fall down to s table,
how many head side will you see?

Many people give a quick answer: 5 head and 5 tails.

Is this the correct answer?

A coin has 2 sides, head and tail side.

If you throw 10 coins up and fall down to s table,
how many head side will you see?

Many people give a quick answer: 5 head and 5 tails.

Is this the correct answer?

It is incorrect!

Why?

A coin has 2 sides, head and tail side.

If you throw 10 coins up and fall down to a table,
how many head side will you see?

Many people give a quick answer: 5 head and 5 tails.

This is incorrect answer. Why?

Correct answer:

"More likely 5 heads and 5 tails. But other cases such as 4 heads or 4 tails are also very likely"

More precisely:

Prob (5 head and 5 tails) =0.246

Thus 5 vs. 5 has prob=0.246

Prob(6 head and 4 tails) = Prob(4 head and 6 tails) = 0.205

Thus 4 vs. 6 has prob=0.405

Prob (7 head and 3 tails) = Prob(3 head and 7 tails) = 0.117

Thus 3 vs. 7 has prob=0.234.

Bernoulli distribution: Simplest probability distribution

Your team win or lose.
You throw a coin; it is head-up or head-down
Your application is accepted or not.

Today is sunny or not-sunny (means rainy or cloudy)
Yow throw a die; the result is 6, or it is not 6 (means 1 or 2 or 3 or 4 or 5)

Bernoulli Distribution

A **Bernoulli distribution** arises from a random experiment which can give rise to just two possible outcomes. These outcomes are usually labeled as either "success" or "failure." If p denotes the probability of a success and the probability of a failure is (1 - p), the the Bernoulli probability function is

$$P(0) = (1-p) \quad and \; P(1) = p$$

Most widely used statistics in real life

Binomial Distribution :

Y = X1 + … + Xn : sum of N independently identically distributed (IID) Bernoulli random variables

One experiment:

- the experiment consists of n independent trials, each with two mutually exclusive outcomes (**success** and **failure**)
- for each trial the probability of success is p (and so the probability of failure is $1 - p$)

Each such trial is called a **Bernoulli trial**.

Experiment: Throwing N identical coins, head-up/head-down

Experiment: Throwing one coin N times

Binomial Distribution Formula

$$P(x) = \binom{n}{x} p^x q^{n-x} = \frac{n!}{(n-x)!\,x!} p^x q^{n-x}$$

where

n = the number of trials (or the number being sampled)

x = the number of successes desired

p = probability of getting a success in one trial

$q = 1 - p$ = the probability of getting a failure in one trial

Example 1

Q. A coin is tossed 10 times. What is the probability of getting exactly 6 heads?

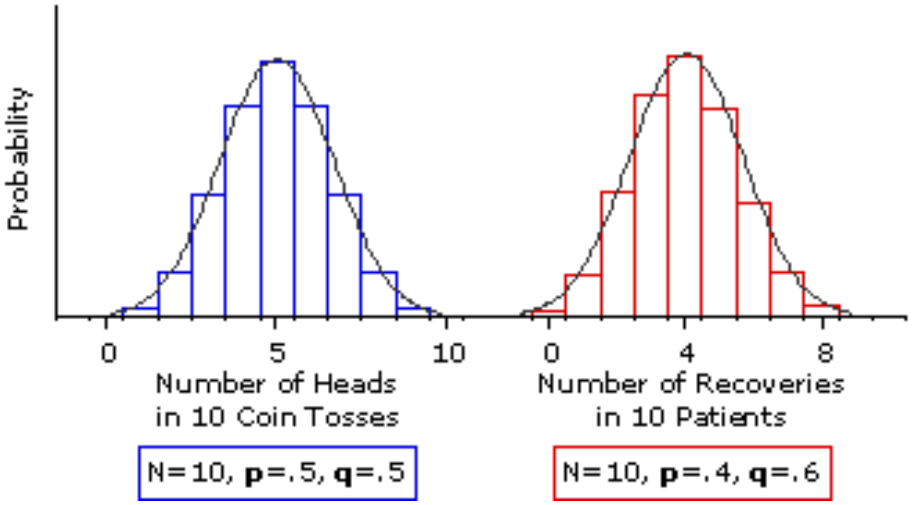
$$p = 0.5,\ q = 1 - p = 0.5,\ n = 10,\ x = 6$$

$$P(x = 6) = \binom{10}{6} 0.5^6\ 0.5^{(10-6)} = 0.2051$$

$$P(x = 5) = 0.2461$$

$$P(x = 3) = P(x = 7) = 0.1172$$

$$P(x = 2) = P(x = 8) = 0.0439$$



Example 3

60% of people who purchase sports cars are men. If 10 sports car owners are randomly selected, find the probability that exactly 7 are men.

$$p = 0.6,\ q = 1 - p = 0.4,\ n = 10,\ x = 7$$

$$P = \binom{10}{7} 0.6^7\ 0.4^{(10-7)} = 0.215$$

How to estimate model parameters

Suppose you don't know p=? How to estimate it.

For binary distribution, simply count (MLE):
p = # of success / total # of occurrence

To estimate the probability for coin headup, you throw 10 coins;
4 head-up, 6 tail-up p=4/(4+6)=0.4

Next time, you throw 8 coins, you get
4 head-up, 4 tail-up p=4/(4+4)=0.5

Another time, you throw 12 coins and you get
7 head-up, 5 tail-up p=7/(7+5)=0.5833

How to compute p ? Average (0.4, 0.5, 0.5833) = 0.4944 ?

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How to compute p ? Average (0.4, 0.5, 0.5833) = 0.4944 ?
No! You do (4+4+7) / (4+6 + 4+4 + 7+5) =15/30=0.5

Bernoulli distribution: two events/outcomes: X=0,1, or success/failure

Multiple distribution: 3 or more events/outcomes: X=1,2,3, …

Example1: X = today's weather: sunny, cloudy, raining

Example2: X (human blood type) = O, A , B, AB

Example3:

A university has undergraduate students and graduate students.
The probability distributions are
p_1 = 0.24 for 1st year students,
p_2 = 0.22 for 2nd year students,
p_3 = 0.20 for 3rd year students,
p_4 = 0.18 for 4th year students,
p_5 = 0.16 for graduate students.

Multinomial Distribution

- The Binomial distribution can be extended to describe number of outcomes in a series of independent trials each having more than 2 possible outcomes.

- If a given trail can result in the k outcomes $E_1, E_2, \ldots, E_k$ with probabilities $p_1, p_2, \ldots, p_k$, then the probability distribution of the random variables $X_1, X_2, \ldots, X_k$, representing the number of occurrences for $E_1, E_2, \ldots, E_k$ in n independent trials is

$$p_{X_1,\ldots,X_k}(x_1,\ldots,x_k) = \frac{n!}{x_1!x_2!\cdots x_k!} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k}$$

with $\sum_{i=1}^{k} x_i = n$, and $\sum_{i=1}^{k} p_i = 1$.

$X_1 = x_1$

……

$X_k = x_k$

Assume a university has undergraduate students and graduate students. The probability distributions are
p_1 = 0.24 for 1st year students,
p_2 = 0.22 for 2nd year students,
p_3 = 0.20 for 3rd year students,
p_4 = 0.18 for 4th year students, and
p_5 = 0.16 for graduate students.

Multinomial Distribution

Then the random vector $X = (X_1, \cdots, X_k)$ is said to be from a multinomial distribution with parameter $(n, p_1, \cdots, p_k)$. We often write

$$X \sim M_k(n; p_1, \cdots, p_k)$$

The multinomial distribution has a nice additive property. Suppose $X \sim M_k(n; p_1, \cdots, p_k)$ and $V \sim M_k(m; p_1, \cdots, p_k)$ and they are independent. It is easy to see that

$$X + V \sim M_k(n + m; p_1, \cdots, p_k).$$

Suppose we focus on one particular category j, then you can easily show that

$$X_j \sim \mathsf{Bin}(n, p_j).$$

Note that $X_1, \cdots, X_k$ are not independent due to the constraint that $X_1 + X_2 + \cdots + X_k = n$. Also, for any X_i and X_j, you can easily show that

$$X_i + X_j \sim \mathsf{Bin}(n, p_i + p_j).$$

An intuitive way to think of this is that the number $X_i + X_j$ is the number of observations in either category i or categoery j. So we are essentially pulling two categories together.

The key idea: additivity:
-- combine the counts and add the numbers of trials
-- the probabilities unchanged.

Example: customers choosing drinks
Suppose each customer independently orders:
- **Coffee** with probability 0.50
- **Tea** with probability 0.30
- **Juice** with probability 0.20

For the first **40 customers**, let X count orders of each drink:
$X \sim \text{Multinomial}(40;0.5,0.3,0.2)$.

For the next **60 customers**, let Ycount orders:
$Y \sim \text{Multinomial}(60;0.5,0.3,0.2)$.

Two groups are independent and have the **same drink probabilities**,
their combined counts satisfy additivity:
$X+Y \sim \text{Multinomial}(100;0.5,0.3,0.2)$.

For example, if the two group of customers are X=(20,12,8),Y=(28,20,12),
then
X+Y=(48,32,20),meaning **48 coffees, 32 teas, and 20 juices** for all 100 customers.

Maximum Likelihood Estimation of model parameters

$$L_n(p_1, \cdots, p_k|X) = \frac{n!}{X_1! \cdots X_k!} p_1^{X_1} \cdots p_k^{X_k}$$

and the log-likelihood function is

$$\ell_n(p_1, \cdots, p_k|X) = \sum_{j=1}^{k} X_j \log p_j + C_n,$$

where C_n a constant is independent of p. Note that naively computing the score function and set it to be 0 will not grant us a solution (think about why) because we do not use the constraint of the parameter space – the parameters are summed to 1. To use this constraint in our analysis, we consider adding the Lagrange multipliers and optimize it:

$$F(p, \lambda) = \sum_{j=1}^{k} X_j \log p_j + \lambda \left(1 - \sum_{j=1}^{k} p_j \right).$$

Differentiating this function with respect to $p_1, \cdots, p_k$, and λ and set it to be 0 gives

$$\frac{\partial F}{\partial p_j} = \frac{X_j}{p_j} - \lambda \Rightarrow X_j = \hat{\lambda} \cdot \hat{p}_{MLE,j}$$

and $1 - \sum_{j=1}^{k} \hat{p}_{MLE,j} = 0$. Thus, $n = \sum_{j=1}^{k} X_j = \hat{\lambda} \sum_{j=1}^{k} p_j = \hat{\lambda}$ so $\boxed{\hat{p}_{MLE,j} = \frac{X_j}{n}}$, which is just the proportion of category j.

simple count

Example:
The distribution of blood types in the US is:

Type	O	A	B	AB
Probability	0.44	0.42	0.10	0.04

In a random sample of 10 Americans, what is the probability 6 have blood type O, 2 have type A, 1 has type B, and 1 has type AB?

$$P(X_1 = x_1, \ldots, X_k = x_k) = \frac{n!}{x_1! \cdots x_k!} p_1^{x_1} \cdots p_k^{x_k}$$

$$P(X_1=6, X_2=2, X_3=1, X_4=1) = \frac{10!}{6!2!1!1!} 0.44^6 0.42^2 0.10^1 0.04^1$$

=0.01290

Homework: MND1

A qualitative assessment. A practical view.

Is P(X1=6, X2=2, X3=1, X4=1) = 0.0129 small?

This seems to be a clueless question!

Hint: Compare to the most probable/likely case!
For 10 persons, X1=4, X2=4, X3=1, X4=1 is the most probable distribution.

P(X1=4, X2=4, X3=1, X4=1) = 0.0294

Thus the most probable (most likely, or maximum) prob = 0.029

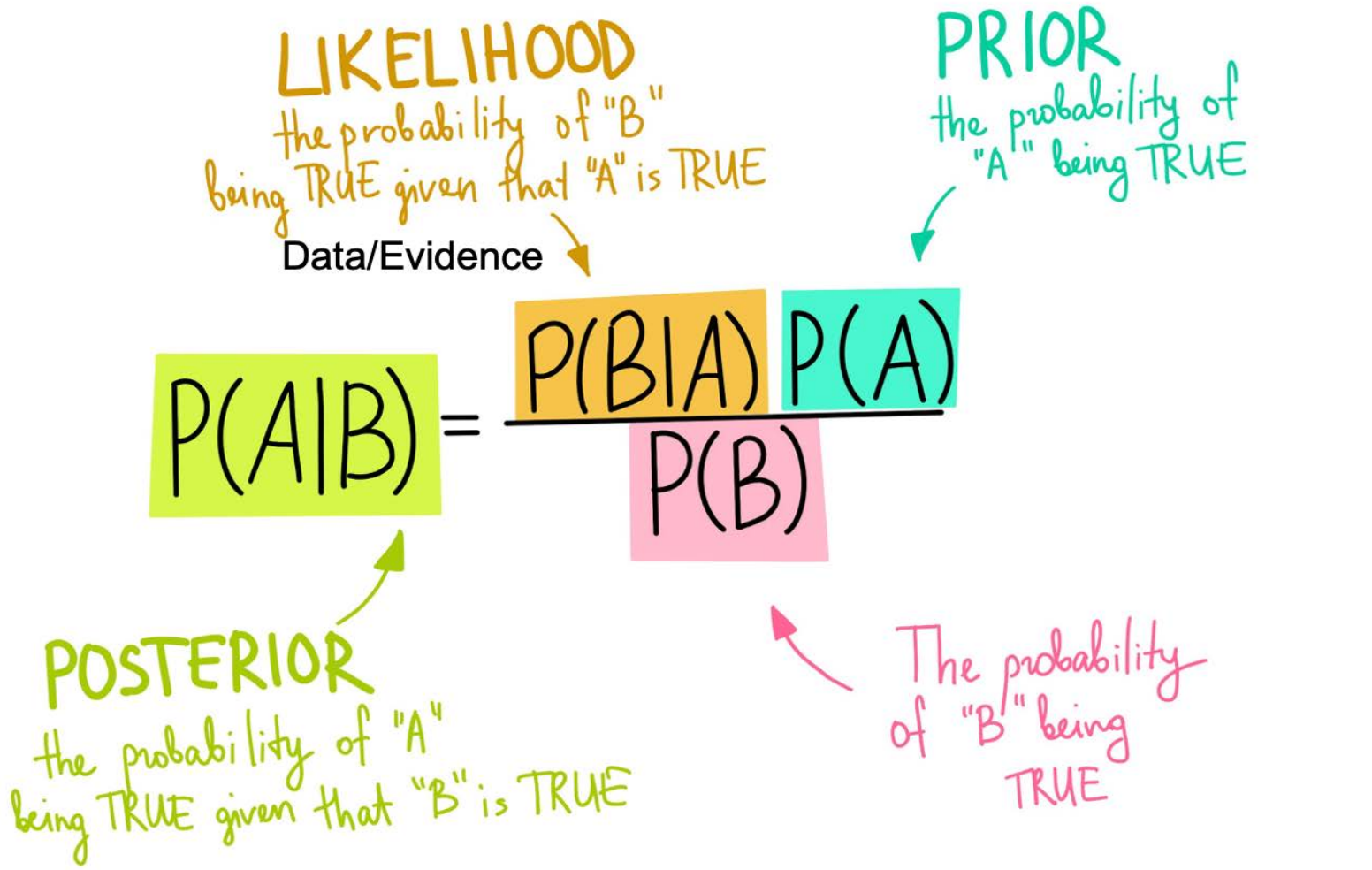
The case of X1=6, X2=2, X3=1, X4=1 is about 0.0129/0.0294= 0.44 of the maximum. Not too small.

Bayes Classification

Using Bayes Theorem (conditional probability) to obtain
the class/label posterior probability of a data instance
given its observed data (attributes/features)

Reading: Textbook Sections 5.3, 5.3.1, 5.3.2, 5.3.3

$$P(A,B) = P(A|B)P(B) = P(B|A)\,P(A)$$



Classification methods

- **Bayes classification**

Bayes Classification for 2 classes

In a perfect world, we have obtained the probability/distribution of every class of data points in data space.

We want to solve the following problem:

In a high school，a student's height is 1.65m. Is the student male or female?

We have the following statistical distributions:
P(male) =70%. Height is Normal dist (mean=1.67m,std=5cm).
P(female) =30%. Height is Normal dist (mean=1.61m,std=5cm).

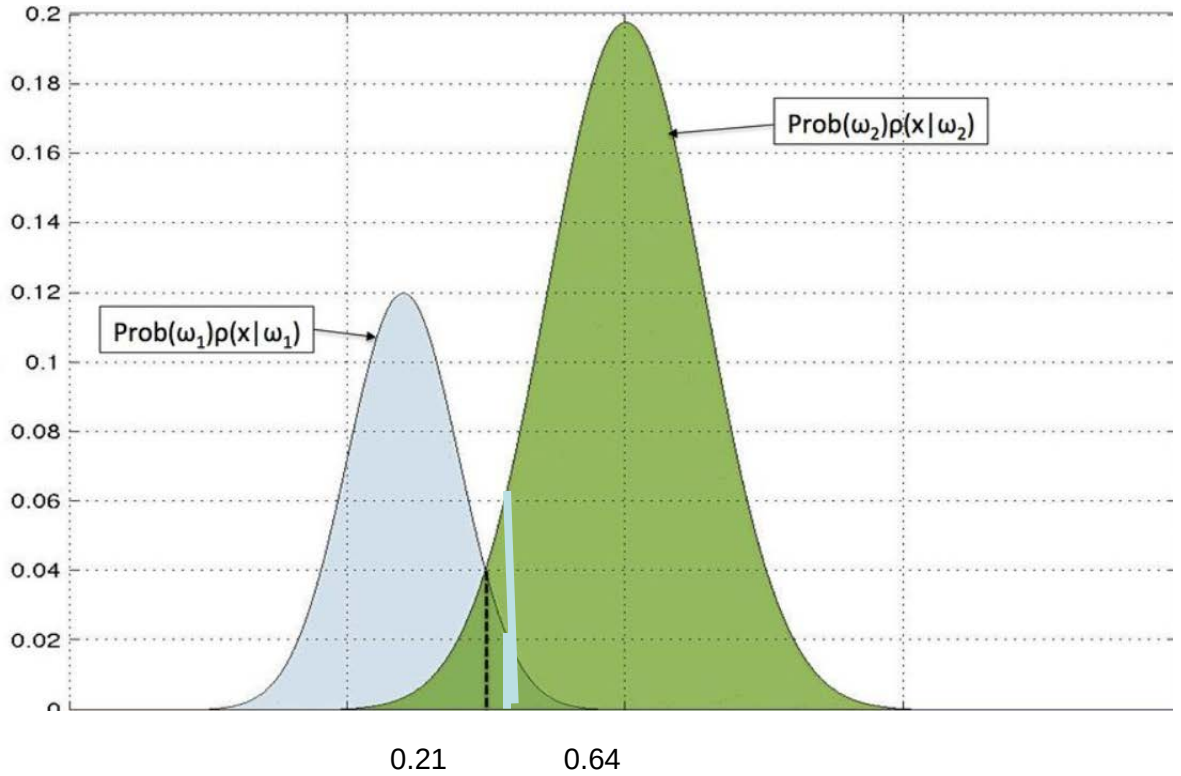
P(male student|x) = P(x=1.65|male)P(male) / P(x=1.65)

P(female student|x) = P(x=1.65|female)P(female) / P(x=1.65)

Bayes Classification for 2 classes

P(male student|x) = P(x=1.65|male)P(male) / P(x=1.65)

P(female student|x) = P(x=1.65|female)P(female) / P(x=1.65)



P(male student | x) = 5.156 / (5.156+1.738)=0.748

P(female student | x) = 1.738 / (5.156+1.738)=0.252

Thus we say this student is male, with an error probability 0.252

This is the best prediction humanly possible. We can not do better than this.

The error is called Bayes error.

Bayes Classification for multiple classes can be done in same fashion.

In a perfect world, we have obtained the probability/distribution of every class of data points in data space.

In a real world, the it is very hard or impossible to obtain the probability/distribution of every class of data points!

Difficulty:

To measure distributions accurately, each dimension requires at least 5 data points. Total minimum data points = 5 ^Dimension.

10-dim requires 5^10 = 9.7 million data points!
20-dim requires 5^20 = $9.5*10^{13}$ data points!

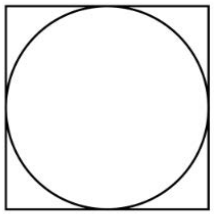
This is called curse of dimensions.

Thus we could not use distribution (density) measures in most situations.

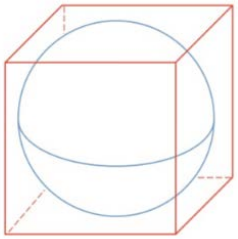
Curse of dimension

hypersphere and a bounding hypercube

2-dim



3-dim



4-dim

5-dim

d-dim

The volume of this hypercube is: $V_{\text{cube}} = (2r)^d$

The volume of a d-dimensional hypersphere is: $V_{\text{sphere}} = \frac{\pi^{d/2}}{\Gamma(\frac{d}{2}+1)} r^d$

$\text{Ratio}(d) = \frac{V_{\text{sphere}}}{} = \frac{\pi^{d/2}}{}$

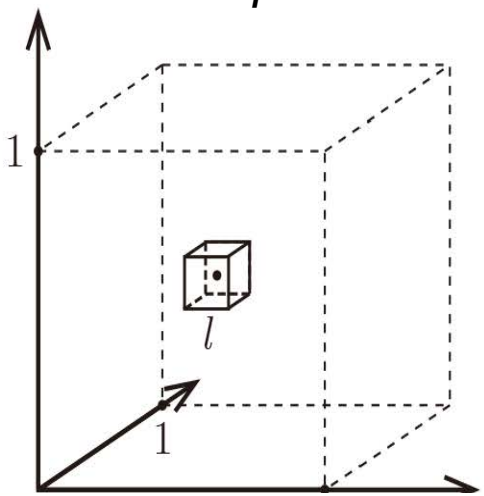
Dimension (d)	Geometry	Ratio Expression	Volume Ratio (%)
1	Line	$2r/2r$	100%
2	Circle in Square	$\pi/4$	~78.5%
3	Sphere in Cube	$\pi/6$	~52.4%
4	Hypersphere	$\pi^2/32$	~30.8%
10	...	...	~0.25%
20	...	...	~0.00002%

Hypersphere shrink …… to zero as dimsion → large

Curse of Dimensionality Intuitive Explanation

Key problem: in high dimensional space, points that are draw from a distribution tends to be far away from each other!

p - dimensional space



$l = s^{1/p}$

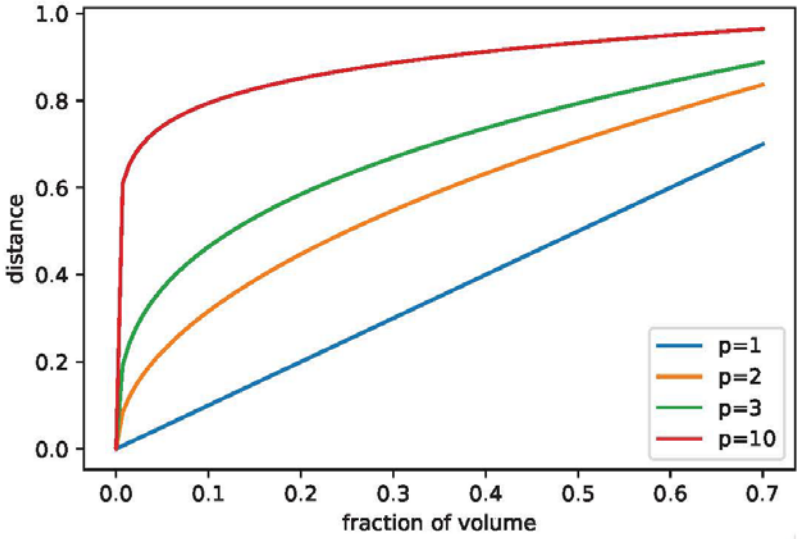
Assume your data lives in $[0, 1]^p$. To capture a neighborhood which represents a fraction s of the hypercube volume, you need the edge length to be $s^{1/p}$

- $s = 0.1,\ p = 10,\ s^{1/p} =$ 0.794
- $s = 0.01,\ p = 10,\ s^{1/p} =$ 0.631

High dimensional spaces are empty

Assume your data lives in $[0,1]^p$. To capture a neighborhood which represents a fraction s of the hypercube volume, you need the edge length to be $s^{1/p}$

- $s = 0.1,\ p = 10,\ s^{1/p} = 0.63$
- $s = 0.01,\ p = 10,\ s^{1/p} = 0.8$



Neighborhoods are no longer local

9

Nearest neighbors

X, Y two independent variables, with uniform distribution on $[0,1]^p$. The mean square distance $\|X-Y\|^2$ satisfies

$$\mathbb{E}[\|X-Y\|^2] = p/6 \text{ and } Std[\|X-Y\|^2] \simeq 0.2\sqrt{p}.$$

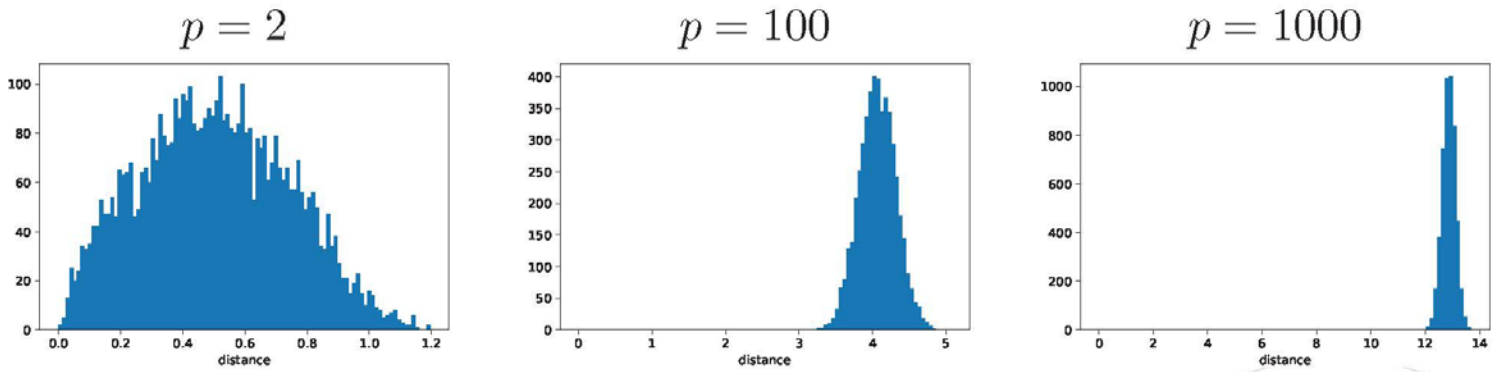


Figure: Histograms of pairwise-distances between $n = 100$ points sampled uniformly in the hypercube $[0,1]^p$

The notion of nearest neighbors vanishes.

A more realistic Bayer Theorem application: Test of Viral Infection

- A medical test for a viral infection. It is 95% reliable for infected patients and 99% reliable for the healthy ones：

- If a patient has the virus (V), and the test shows (S = test show virus) with probability $P(S \mid V) = 0.95.$

- If a patient does not have the virus, the test confirms that with probability $P(\bar{S} \mid \bar{V}) = 0.99.$

- A patient tests positive (the test shows that the patient has the virus).
- Does this means he has 95% probability of the virus?
- No!
- Because the question refers the probability that his test shows he has virus, we ask the prob that he indeed has virus, i.e., P{V|S} . This is not given directly in the statement of the problem.
- We compute P{V|S} using Bayes theorem.

Bayes' Rule

- Bayes Theorem (conditional probability):

$$P\{B \mid A\} = \frac{P\{A \mid B\}P\{B\}}{P\{A\}} = \frac{P\{A \mid B\}P\{B\}}{P\{A \mid B\}P\{B\} + P\{A \mid \overline{B}\}P\{\overline{B}\}}$$

Law of Total Probability

$$P\{A\} = \sum_{j=1}^{k} p\{\mathrm{A} \mid B_j\} P\{B_j\}$$

In case of two events (k=2),

$$P\{A\} = P\{A \mid B\} P\{B\} + P\{A \mid \overline{B}\} P\{\overline{B}\}$$

Medical Test Example cont.

- We need additional information: Suppose 4% of all the population are infected with the virus, P{V} = 0.04.
- Recall: $P\{S \mid V\} = 0.95 \quad P\{\overline{S} \mid \overline{V}\} = 0.99$

- The desired (conditional) probability is

$$P\{V \mid S\} = \frac{P\{S \mid V\}P\{V\}}{P\{S \mid V\}P\{V\} + P\{S \mid \overline{V}\}P\{\overline{V}\}}$$

$$= \frac{(0.95)(0.04)}{(0.95)(0.04) + (1-0.99)(1-0.04)} = 0.7983$$

Test of Viral Infection - Conclusion

- Thus the probability of the patient has the virus is 79.83%, not 95%.
- Bayesian view:

  This patient has 4% probability of been infected by the virus [because 4% of the population has the virus]. Because now he tested positive for the virus, his chance of virus increased to 79.83%.

  This patient has 4% probability of been infected by the virus [because 4% of the population has the virus (prior probability)]. Because now he tested positive for the virus (new data evidence), his chance of virus increased to 79.83%.

Machine Learning Rule/Principle No.1:

- We use the simplest models, distributions to describe the data

Why?

Because it has the least over-fitting, thus generalize to other data better.

For continuous data, we use Gaussian distribution.

For discrete data, we use multinomial distribution.

For fitting data, we mostly use linear regression.

Etc.

(If we have detailed domain knowledge of the problem, we may incorporate that knowledge into the analysis model)

Naïve Bayes Classification

Using Bayes Theorem (conditional probability) to obtain
the class/label posterior probability of a data instance
given its observed data (attributes/features)

In 1970s, scientists began to do Language /text analysis. They need to compute:

Prob("I am happy because at lunch, I ate some very good fish") = ?

Prob("Oh God! The road is too congested")=?

Prob("It's horrible. I don't believe it")=?

This task seems to be clueless, impossible to do.

Naïve Bayes come to the rescue!

We need to compute the probability of occurrence of
"I am happy because at lunch, I ate some very good fish"

We don't really know how to compute

Prob("I am happy because at lunch, I ate some very good fish") = ?

But naïve Bayes assumes we can do

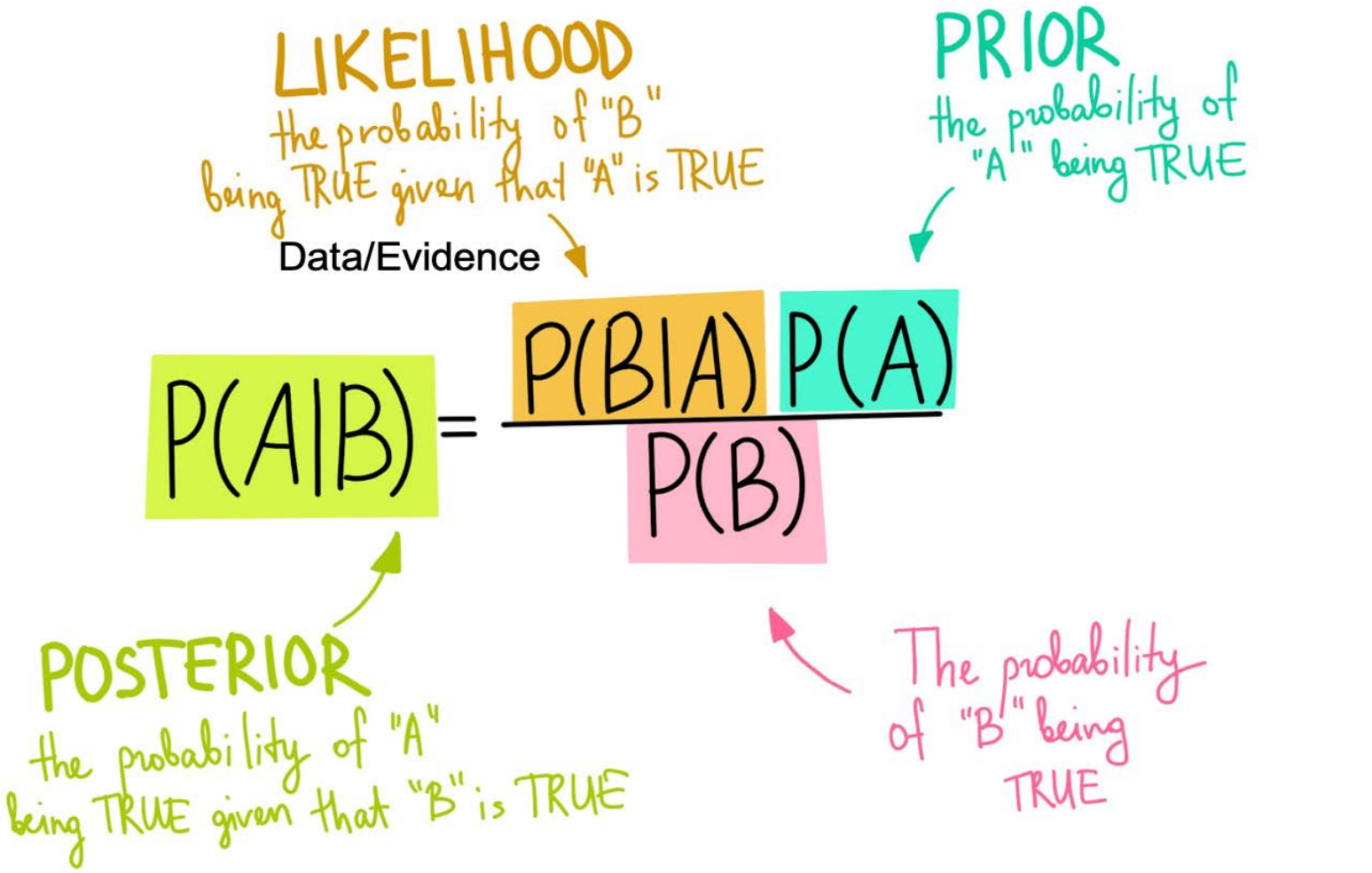
Prob("I am happy because at lunch, I ate some very good fish")
=
Prob("I"） * Prob("am") * Prob("happy") … Prob("good") * Prob("fish")

We compute each of them assuming word distributed following the multinomial distribution.

Thus naïve Bayes method makes **something impossible** to possible !

In practice, Naïve Bayes assumes that
given class, words distribute independently

$P(A,B) = P(A|B)P(B) = P(B|A)\,P(A)$



5.3.3 Naïve Bayes Classifier

A naïve Bayes classifier estimates the class-conditional probability by assuming that the attributes are conditionally independent, given the class label y. The conditional independence assumption can be formally stated as follows:

(the simplification)

$$P(\mathbf{X}|Y = y) = \prod_{i=1}^{d} P(X_i|Y = y), \tag{5.12}$$

where each attribute set $\mathbf{X} = \{X_1, X_2, \ldots, X_d\}$ consists of d attributes.

Probability of occurrence of X is equal to the product of the probability of occurrence of every attributes of X given the class of X

This says each class has a different multinomial distribution of attributes.

To classify a test record, the naïve Bayes classifier computes the posterior probability for each class Y:

$$P(Y|\mathbf{X}) = \frac{P(Y)\prod_{i=1}^{d} P(X_i|Y)}{P(\mathbf{X})}. \quad (5.15)$$

Since $P(\mathbf{X})$ is fixed for every Y, it is sufficient to choose the class that maximizes the numerator term, $P(Y)\prod_{i=1}^{d} P(X_i|Y)$.

P(w_1, w_2, …w_d | Y) = P(w_1|Y) *P(w_2|Y)* … * P(w_d|Y)

the prior probability $P(Y)$

the class-conditional probabilities $\prod_i P(X_i|Y)$, = multinomial distribution of attributes for class Y

Loan Applications of a Bank

Feature/attribute/variable

class

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Loan borrower not able to pay required monthly payments

class

P(Home Owner=Yes|No) = 3/7
P(Home Owner=No|No) = 4/7
P(Home Owner=Yes|Yes) = 0
P(Home Owner=No|Yes) = 1
P(Marital Status=Single|No) = 2/7
P(Marital Status=Divorced|No) = 1/7
P(Marital Status=Married|No) = 4/7
P(Marital Status=Single|Yes) = 2/3
P(Marital Status=Divorced|Yes) = 1/3
P(Marital Status=Married|Yes) = 0

Compute probability of occurrence of each attributes for class Y="no"

Compute probability of occurrence of each attributes for class Y="yes"

Attribute income is binned, and becomes an ordinal attribute = low if < 80K, middle if between 80K and 101K, high if above 101K

Loan Applications of a Bank

Naïve Bayes: Class-conditional distribution for attribute: home-owner

Tid	Home Owner	Defaulted Borrower
1	Yes	No
2	No	No
3	No	No
4	Yes	No
5	No	No
6	No	No
7	Yes	No
8	No	Yes
9	No	Yes
10	No	Yes

$$P(\text{Home Owner} = \text{Yes} \mid \text{Class} = \text{No}) = 3/7$$

$$P(\text{Home Owner} = \text{No} \mid \text{Class} = \text{No}) = 4/7$$

$$P(\text{Home Owner} = \text{Yes} \mid \text{Class} = \text{Yes}) = 0 \quad = 0/3$$

$$P(\text{Home Owner} = \text{No} \mid \text{Class} = \text{Yes}) = 1 \quad = 3/3$$

Compute probability of occurrence of each attributes for class Y="no"

Compute probability of occurrence of each attributes for class Y="yes"

Loan Applications of a Bank

Naïve Bayes: Class-conditional distribution for attribute: marital status

Tid	Marital Status	Defaulted Borrower
1	Single	No
2	Married	No
3	Single	No
4	Married	No
5	Married	No
6	Married	No
7	Divorced	No
8	Single	Yes
9	Divorced	Yes
10	Single	Yes

$$P(\text{Marital Status} = \text{Single} \mid \text{Class} = \text{No}) = 2/7$$

$$P(\text{Marital Status} = \text{Married} \mid \text{Class} = \text{No}) = 1/7$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{Class} = \text{No}) = 4/7$$

$$P(\text{Marital Status} = \text{Single} \mid \text{Class} = \text{yes}) = 2/3$$

$$P(\text{Marital Status} = \text{Married} \mid \text{Class} = \text{yes}) = 1/3$$

$$P(\text{Marital Status} = \text{Divorced} \mid \text{Class} = \text{yes}) = 0$$

Compute probability of occurrence of each attributes for class Y="no"

Compute probability of occurrence of each attributes for class Y="yes"

Loan Applications of a Bank

Naïve Bayes: Class-conditional distribution for attribute: annual income

Tid	Annual Income	Defaulted Borrower
1	125K	No
2	100K	No
3	70K	No
4	120K	No
5	75K	No
6	60K	No
7	220K	No
8	85K	Yes
9	95K	Yes
10	90K	Yes

$$P(\text{Annual Income} = \text{Low} \mid \text{Class} = \text{No}) = 3/7$$

$$P(\text{Annual Income} = \text{Middle} \mid \text{Class} = \text{No}) = 1/7$$

$$P(\text{Annual Income} = \text{High} \mid \text{Class} = \text{No}) = 3/7$$

$$P(\text{Annual Income} = \text{Low} \mid \text{Class} = \text{Yes}) = 0$$

$$P(\text{Annual Income} = \text{Middle} \mid \text{Class} = \text{Yes}) = 1$$

$$P(\text{Annual Income} = \text{High} \mid \text{Class} = \text{Yes}) = 0$$

Compute probability of occurrence of each attributes for class Y="no"

Compute probability of occurrence of each attributes for class Y="yes"

Attribute income is binned, and becomes an ordinal attribute = low if < 80K, middle if between 80K and 101K, high if above 101K

Standard multinomial distribution parameter estimation:

$$P(x_i|Y=y)^{\text{MLE}} = p_{i,y}^{\text{MLE}} = \frac{n_{i,y}}{N_y}$$

where $n_{i,y}$ is the number of training examples in class y where attribute x_i occurs, N_y is the number of training examples in class y.

Laplace smoothed multinomial distribution parameter estimation:

See 2nd Edition Textbook p.224

$$P(x_i|Y=y)^{\text{smoothed}} = p_{i,y}^{\text{smoothed}} = \frac{n_{i,y}+1}{N_y+\nu_y}$$

where ν is the total number of attributes in class y.

In most applications, we use Laplace smoothed parameter estimation

Mammals Data using Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

4 attributes
2 classes: (M,N)

-- Need to compute 8
class probabilities
for attributes.
-- Need to compute class prior

-- Compute P(M|X), P(NM|X)

X =

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

Mammals Data using Naïve Bayes Classifier

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

A: attributes

M: mammals

N: non-mammals

$$
\begin{aligned}
&P(A \mid M) \\
&= P(Give\ Birth = yes \mid Class = M) \\
&\times P(Can\ Fly = no \mid Class = M) \\
&\times P(Live\ in\ Water = yes \mid Class = M) \\
&\times P(Have\ Legs = no \mid Class = M) \\
&= \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06
\end{aligned}
$$

x =

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

(A) compute P(M | X)=P(X|M)*P(M)/P(X), P(N | X)=P(X|N)*P(N)/P(X),
(B) assign x to the class with higher probability